MANIPAL INSTITUTE OF TECHNOLOGY

Manipal Academy of Higher Education, Manipal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THIRD SEMESTER B.TECH. END SEMESTER EXAMINATION - 2025

CS 2101: DATA STRUCTURES

Branch: Computer Science & Engineering (CSE) | Credit: 4

Time: 3 Hours | Max. Marks: 50

Instructions: Answer ALL questions. Each question carries designated marks.

PART A

1. (a) Analyze Circular Queue implementation using fixed-size array and boundary overflow conditions. [5 Marks]

1. (b) Construct an AVL tree for insertions: [21, 26, 30, 9, 4, 14, 28, 18, 15] showing rotations. [5 Marks]

PART B

2. (a) Explain Min-Heap building algorithm and HeapSort in-place reordering time complexity. [5 Marks]

2. (b) Write Kruskal minimum spanning tree algorithm with Disjoint Set Union (DSU) path compression. [5 Marks]

PART C

3. (a) Differentiate Open Addressing (Linear, Quadratic) vs Chaining in Hash Tables with collision rates. [10 Marks]

PART D

4. (a) Formulate and solve recurrence relations using Master Theorem for Divide-and-Conquer. [10 Marks]

[END OF QUESTION PAPER - MIT MANIPAL CSE 3RD SEMESTER ARCHIVE]